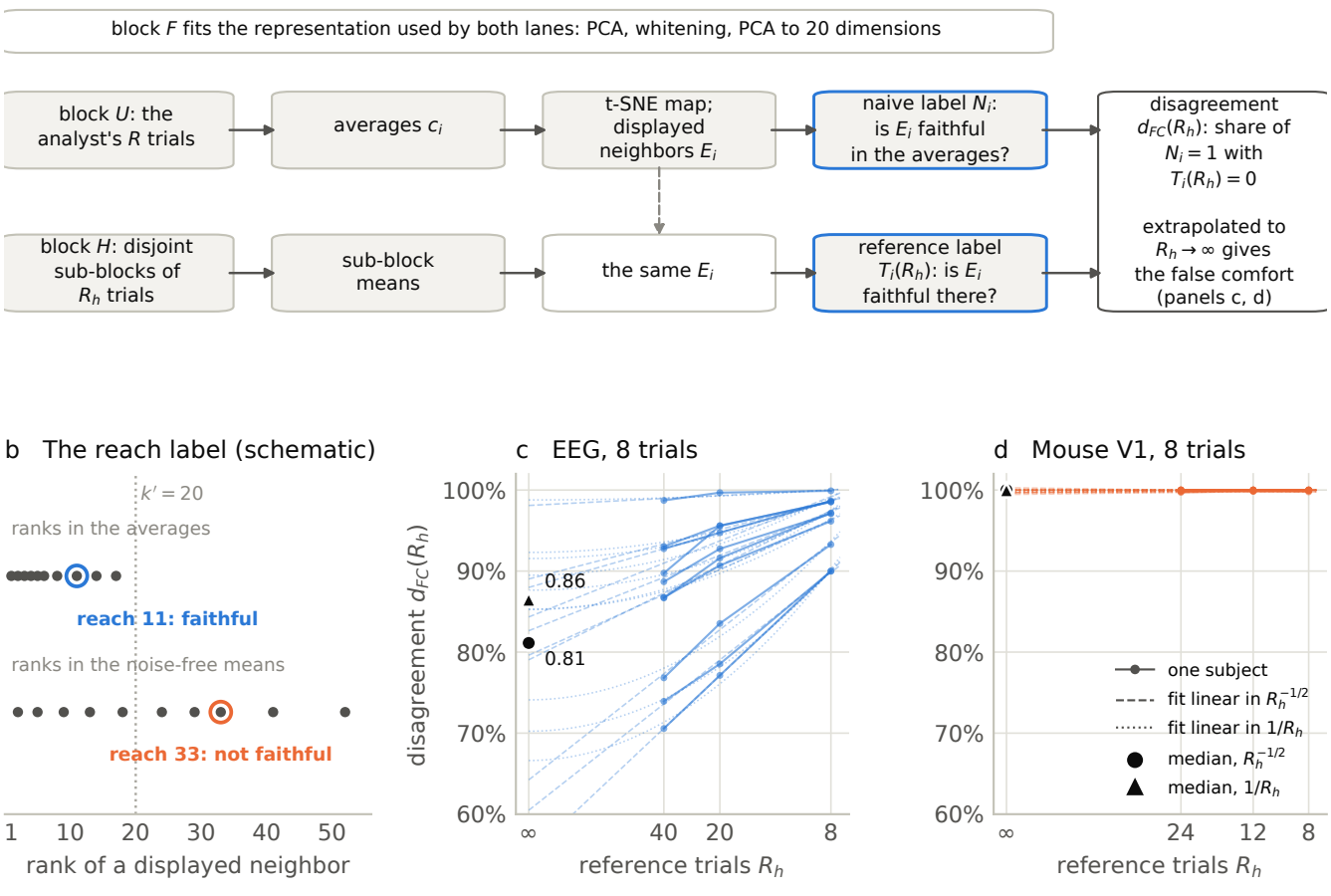

a The estimator, for each subject and random split of the repetitions
block F fits the representation used by both lanes: PCA, whitening, PCA to 20 dimensions
block U: the analyst's R trials
averages c_i
t-SNE map; displayed neighbors E_i
naive label N_i: is E_i faithful in the averages?
disagreement $d_{FC}(R_h)$: share of $N_i = 1$ with $T_i(R_h) = 0$
extrapolated to $R_h \to \infty$ gives the false comfort (panels c, d)
block H: disjoint sub-blocks of R_h trials
sub-block means
the same E_i
reference label $T_i(R_h)$: is E_i faithful there?
b The reach label (schematic)
$k' = 20$
ranks in the averages
reach 11: faithful
ranks in the noise-free means
reach 33: not faithful
1 10 20 30 40 50
rank of a displayed neighbor
c EEG, 8 trials
disagreement $d_{FC}(R_h)$
100%
90%
80%
70%
60%
0.86
0.81
∞ 40 20 8
reference trials R_h
d Mouse V1, 8 trials
one subject
fit linear in $R_h^{-1/2}$
fit linear in $1/R_h$
median, $R_h^{-1/2}$
median, $1/R_h$
∞ 24 12 8
reference trials R_h